

Complete Session, Presentation & Application Layer Revision Hinglish Readme

Computer Networks — Session Layer, Presentation Layer & Application Layer Complete Revision Notes (Hinglish)

UNIT: SESSION LAYER

1. Session Layer

OSI Model ki 5th layer.

Ye layer sender aur receiver ke beech communication session ko establish, maintain aur terminate karti hai.

Main Functions of Session Layer

1 Session Establishment

Communication start karna.

Example

Browser aur web server ke beech connection establish karna.

2 Session Maintenance

Connection ko active rakhna.

Agar temporary interruption aaye toh session ko manage karta hai.

3 Session Termination

Communication complete hone par session close karta hai.

4 Dialog Control

Kaun device kab data bhejega ye decide karta hai.

Communication Modes

Mode	Meaning
Half Duplex	Ek time pe ek side data bhej sakta
Full Duplex	Dono side same time data bhej sakte

Example

- Walkie Talkie → Half Duplex
 - Mobile Call → Full Duplex
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5 Synchronization

Large data transfer me checkpoints banana.

Agar transmission fail ho jaye toh wahi se continue ho sakta hai.

Session Layer Protocols

Protocol	Work
NetBIOS	LAN communication
PPTP	VPN sessions
RPC	Remote procedure calls

UNIT: PRESENTATION LAYER

2. Presentation Layer

OSI Model ki 6th layer.

Data ko readable format me convert karti hai aur security provide karti hai.

Main Functions

Function	Work
Translation	Data format convert
Encryption	Data secure
Compression	Data size reduce

3. Cryptography

Data ko secure karne ki technique.

Important Terms

Term	Meaning
Plaintext	Original message
Ciphertext	Encrypted message
Encryption	Plain → Cipher
Decryption	Cipher → Plain
Key	Secret value

4. Symmetric Key Cryptography

Definition

Encryption aur decryption dono me SAME key use hoti hai.

Working

Plaintext + Secret Key → Ciphertext
Ciphertext + Same Key → Plaintext

Advantages

- Fast
- Less computation
- Large data encryption ke liye best

Disadvantages

- Key sharing problem
- Key leak hone par security risk

Examples

- AES
- DES
- Blowfish

5. Asymmetric Key Cryptography

Definition

Isme 2 keys use hoti hain:

- Public Key
- Private Key

Working

Public Key → Encryption
Private Key → Decryption

Advantages

- More secure
 - Key sharing issue nahi
-

Disadvantages

- Slow
 - More processing power
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Examples

- RSA
 - ECC
 - Diffie-Hellman
-

Symmetric vs Asymmetric

Feature	Symmetric	Asymmetric
Keys	1	2
Speed	Fast	Slow
Security	Lower	Higher
Usage	Bulk encryption	Secure communication

6. Digital Signature

Definition

Electronic signature jo verify karti hai:

- Sender original hai
 - Message modify nahi hua
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Working

Sender Side

1. Message hash create hota hai

2. Hash private key se encrypt hota hai
 3. Digital signature generate hota hai
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Receiver Side

1. Public key se decrypt
 2. Hash compare
 3. Same hua toh message authentic
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Benefits

- Authentication
 - Integrity
 - Non-repudiation
-

7. Compression

Data ka size reduce karna.

Types of Compression

1 Lossless Compression

Original data perfectly recover ho sakta hai.

Examples

- ZIP
- PNG
- GIF

Advantages

- No data loss

Disadvantages

- Compression ratio low
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2 Lossy Compression

Thoda data permanently remove hota hai.

Examples

- JPEG
- MP3
- MP4

Advantages

- High compression

Disadvantages

- Quality reduce hoti hai

Lossless vs Lossy

Feature	Lossless	Lossy
Data Loss	No	Yes
Quality	Same	Reduced
Compression	Lower	Higher

UNIT: APPLICATION LAYER

8. Application Layer

OSI Model ki 7th layer.

Ye user ke closest layer hoti hai.

Applications ko network services provide karti hai.

Main Protocols

- DNS
- HTTP
- FTP

- SMTP
 - SNMP
 - Telnet
-

9. Domain Name System (DNS)

Definition

DNS internet ka phonebook hota hai.

Domain name ko IP address me convert karta hai.

Example

google.com → 142.xxx.xxx.xxx

DNS Working

1. Browser domain request karta hai
 2. DNS server IP provide karta hai
 3. Browser server connect karta hai
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10. World Wide Web (WWW)

Interconnected web pages ka collection.

WWW uses:

- HTTP
 - Web browsers
 - Hyperlinks
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11. Hyper Text Transfer Protocol (HTTP)

Browser aur server ke beech communication protocol.

Features of HTTP

- Stateless
 - Request-response based
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HTTP Methods

Method	Work
GET	Data retrieve
POST	Data send
PUT	Update data
DELETE	Remove data

12. Electronic Mail (E-Mail)

Email transfer ke protocols.

E-Mail Protocols

Protocol	Work
SMTP	Send mail
POP3	Receive and download
IMAP	Receive and sync

E-Mail Working

1. Sender mail bhejta hai
 2. SMTP server transfer karta hai
 3. Receiver POP3/IMAP se receive karta hai
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13. File Transfer Protocol (FTP)

Files transfer karne ka protocol.

Uses of FTP

- Upload files
 - Download files
 - Website hosting
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FTP Ports

Port	Work
20	Data transfer
21	Control connection

14. Simple Network Management Protocol (SNMP)

Network devices monitor aur manage karne ka protocol.

Uses

- Router monitoring
 - Switch monitoring
 - Traffic analysis
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Components of SNMP

Component	Work
Manager	Controls network
Agent	Device software
MIB	Database

15. Telnet

Remote login protocol.

Ek computer ko remotely access karne ke liye use hota hai.

Features

- Remote command execution
 - Plaintext communication
 - Secure nahi hota
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Port Number

Port 23

Telnet Limitation

Data plaintext me send hota hai.

Isliye SSH zyada secure alternative hai.

QUICK REVISION TABLE

Topic	Main Work
Session Layer	Session management
Presentation Layer	Encryption + Compression
Symmetric Crypto	Same key encryption
Asymmetric Crypto	Public/private keys
Digital Signature	Authentication
Lossless Compression	No data loss
Lossy Compression	High compression
DNS	Domain → IP
HTTP	Web communication
FTP	File transfer
SMTP	Send mail
SNMP	Network monitoring

Topic

Main Work

Telnet

Remote login

MEMORY TRICKS

Symmetric Key

"Same key both sides"

Asymmetric Key

"Public lock, private unlock"

DNS

"Internet phonebook"

SMTP

"Send Mail To People"

MIND MAP

OSI Upper Layers



